

Round Robin CPU Scheduling Algorithm

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

Input Format

- The first line of input contains an integer representing the number of processes.
- The second line contains an integer representing the Time Quantum.
- The next lines contain two integers for each process, where
 - The first integer denotes the Arrival Time (AT)
 - The second integer denotes the Burst Time (BT) of the process.

Output Format

The output displays the Average Turnaround Time and the Average Waiting Time.

Sample Input 1

```
3
2
0 5
1 3
2 1
```

Sample Output 1

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
```

```
1 #include <stdio.h>
2 int main(){
3     int n, tq, i;
4     printf("Enter number of processes: \n");
5     scanf("%d", &n);
6     printf("Enter Time Quantum: \n");
7     scanf("%d", &tq);
8     int at[n], bt[n], ct[n], tat[n], wt[n], rt[n];
9     int completed = 0, time = 0;
10    for(i = 0; i < n; i++){
11        printf("Enter AT and BT for P%d: \n", i+1);
12        scanf("%d %d", &at[i], &bt[i]);
13        rt[i] = bt[i];
14    }
15    float totaltat=0, totalwt=0;
16    while(completed<n){
17        int executed=0;
18        for(i=0;i<n;i++){
19            if(at[i]<time&&rt[i]>0){
20                executed=1;
21                if(rt[i]>tq){
22                    time+=tq;
23                    rt[i]-=tq;
24                }
25                else{
26                    time+=rt[i];
27                    rt[i]=0;
28                    ct[i]=time;
29                    completed++;
30                }
31            }
32        }
33        if(!executed){
34            time++;
35        }
36    }
37    for(i=0;i<n;i++){
38        tat[i]=ct[i]-at[i];
39        wt[i]=tat[i]-bt[i];
40        totaltat+=tat[i];
41        totalwt+=wt[i];
42    }
43    printf("\nAverage Turnaround Time: %.2f\n",totaltat/n);
44    printf("Average Waiting Time: %.2f\n",totalwt/n);
45    return 0;
46 }
```

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Output Format

The output displays the Average Turnaround Time and the Average

Custom Test Case

Custom Test

Success ✓

Input :

```
3
2
0 5
1 3
2 1
```

Expected Output :

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
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Output :

```
Enter number of processes:
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